

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import string
import nltk
```

```
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\gurug\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]   C:\Users\gurug\AppData\Roaming\nltk_data...
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```

```
True
```

```
df = pd.read_excel("amazon.xlsx")
```

```
df.head()
```

	Star	HelpFul	Title \
0	5	0	looks great
1	5	0	Pattern did not align between the two panels.
2	5	0	Imagery is stretched. Still fun.
3	5	0	Que se ven elegantes muy finas
4	5	0	Wow great purchase

	Review
0	Happy with it
1	Good quality material however the panels are m...
2	Product was fun for bedroom windows. Imag...
3	Lo unico que me gustaria es que sean un poco ...
4	Great bang for the buck I can't believe the qu...

```

print(df.shape)

df.info()

df.isnull().sum()

(5611, 4)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5611 entries, 0 to 5610
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0    Star        5611 non-null    int64
1    HelpFul     5611 non-null    int64
2    Title       5559 non-null    object
3    Review      5593 non-null    object
dtypes: int64(2), object(2)
memory usage: 175.5+ KB

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0    Star        5611 non-null    int64
1    HelpFul     5611 non-null    int64
2    Title       5559 non-null    object
3    Review      5593 non-null    object
dtypes: int64(2), object(2)
memory usage: 175.5+ KB

Star        0
HelpFul     0
Title       52
Review      18
dtype: int64

df = df[['Star', 'Review']]

df.head()

   Star      Review
0     5  Happy with it
1     5  Good quality material however the panels are m...
2     5  Product was fun for bedroom windows.<br />Imag...
3     5  Lo unico que me gustaria es que sean un poco ...
4     5  Great bang for the buck I can't believe the qu...

df = df[df['Star'] != 3]

```

```
df['label'] = df['Star'].apply(lambda x: 1 if x >= 4 else 0)
```

```
df.head()
```

	Star	Review	label
0	5	Happy with it	1
1	5	Good quality material however the panels are m...	1
2	5	Product was fun for bedroom windows. Imag...	1
3	5	Lo unico que me gustaria es que sean un poco ...	1
4	5	Great bang for the buck I can't believe the qu...	1

```
df["clean_text"] = df["Review"].astype(str).str.lower()
```

```
df.head()
```

	Star	Review	label	\
0	5	Happy with it	1	
1	5	Good quality material however the panels are m...	1	
2	5	Product was fun for bedroom windows. Imag...	1	
3	5	Lo unico que me gustaria es que sean un poco ...	1	
4	5	Great bang for the buck I can't believe the qu...	1	

	clean_text
0	happy with it
1	good quality material however the panels are m...
2	product was fun for bedroom windows. imag...
3	lo unico que me gustaria es que sean un poco ...
4	great bang for the buck i can't believe the qu...

```
df["clean_text"] =  
df["clean_text"].str.replace(f"[{string.punctuation}]", "",  
regex=True)
```

```
df.head()
```

	Star	Review	label	\
0	5	Happy with it	1	
1	5	Good quality material however the panels are m...	1	
2	5	Product was fun for bedroom windows. Imag...	1	
3	5	Lo unico que me gustaria es que sean un poco ...	1	
4	5	Great bang for the buck I can't believe the qu...	1	

	clean_text
0	happy with it
1	good quality material however the panels are m...
2	product was fun for bedroom windowsbr imagery ...
3	lo unico que me gustaria es que sean un poco ...
4	great bang for the buck i cant believe the qua...

```

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer

```

```

df["tokens"] = df["clean_text"].apply(word_tokenize)

```

```

df.head()

```

	Star	Review	label	\
0	5	Happy with it	1	
1	5	Good quality material however the panels are m...	1	
2	5	Product was fun for bedroom windows. Imag...	1	
3	5	Lo unico que me gustaria es que sean un poco ...	1	
4	5	Great bang for the buck I can't believe the qu...	1	

	clean_text	\
0	happy with it	
1	good quality material however the panels are m...	
2	product was fun for bedroom windowsbr imagery ...	
3	lo unico que me gustaria es que sean un poco ...	
4	great bang for the buck i cant believe the qua...	

	tokens
0	[happy, with, it]
1	[good, quality, material, however, the, panels...
2	[product, was, fun, for, bedroom, windowsbr, i...
3	[lo, unico, que, me, gustaria, es, que, sean, ...
4	[great, bang, for, the, buck, i, cant, believe...

```

stop_words = set(stopwords.words("english"))

```

```

df["tokens"] = df["tokens"].apply(
    lambda words: [word for word in words if word not in stop_words]
)

```

```

df.head()

```

	Star	Review	label	\
0	5	Happy with it	1	
1	5	Good quality material however the panels are m...	1	
2	5	Product was fun for bedroom windows. Imag...	1	
3	5	Lo unico que me gustaria es que sean un poco ...	1	
4	5	Great bang for the buck I can't believe the qu...	1	

	clean_text	\
0	happy with it	
1	good quality material however the panels are m...	
2	product was fun for bedroom windowsbr imagery ...	
3	lo unico que me gustaria es que sean un poco ...	
4	great bang for the buck i cant believe the qua...	

```

                                tokens
0                                [happy]
1  [good, quality, material, however, panels, mis...
2  [product, fun, bedroom, windowsbr, imagery, bi...
3  [lo, unico, que, gustaria, es, que, sean, un, ...
4  [great, bang, buck, cant, believe, quality, ma...

stop_words = set(stopwords.words("english"))

df["tokens"] = df["tokens"].apply(
    lambda words: [word for word in words if word not in stop_words]
)

df.head()

```

	Star	Review	label	\
0	5	Happy with it	1	
1	5	Good quality material however the panels are m...	1	
2	5	Product was fun for bedroom windows. Imag...	1	
3	5	Lo unico que me gustaria es que sean un poco ...	1	
4	5	Great bang for the buck I can't believe the qu...	1	

```

                                clean_text \
0                                happy with it
1  good quality material however the panels are m...
2  product was fun for bedroom windowsbr imagery ...
3  lo unico que me gustaria es que sean un poco ...
4  great bang for the buck i cant believe the qua...

```

```

                                tokens
0                                [happy]
1  [good, quality, material, however, panels, mis...
2  [product, fun, bedroom, windowsbr, imagery, bi...
3  [lo, unico, que, gustaria, es, que, sean, un, ...
4  [great, bang, buck, cant, believe, quality, ma...

lemmatizer = WordNetLemmatizer()

df["lemma"] = df["tokens"].apply(
    lambda words: [lemmatizer.lemmatize(word) for word in words]
)

df.head()

```

	Star	Review	label	\
0	5	Happy with it	1	
1	5	Good quality material however the panels are m...	1	
2	5	Product was fun for bedroom windows. Imag...	1	
3	5	Lo unico que me gustaria es que sean un poco ...	1	
4	5	Great bang for the buck I can't believe the qu...	1	

```

                                clean_text \
0                                happy with it
1 good quality material however the panels are m...
2 product was fun for bedroom windowsbr imagery ...
3 lo unico que me gustaria es que sean un poco ...
4 great bang for the buck i cant believe the qua...

```

```

                                tokens \
0                                [happy]
1 [good, quality, material, however, panels, mis...
2 [product, fun, bedroom, windowsbr, imagery, bi...
3 [lo, unico, que, gustaria, es, que, sean, un, ...
4 [great, bang, buck, cant, believe, quality, ma...

```

```

                                lemma
0                                [happy]
1 [good, quality, material, however, panel, mism...
2 [product, fun, bedroom, windowsbr, imagery, bi...
3 [lo, unico, que, gustaria, e, que, sean, un, p...
4 [great, bang, buck, cant, believe, quality, ma...

```

```

df["preprocessed_text"] = df["lemma"].apply(lambda words: "
".join(words))

```

```

df.head()

```

	Star	Review	label
0	5	Happy with it	1
1	5	Good quality material however the panels are m...	1
2	5	Product was fun for bedroom windows. Imag...	1
3	5	Lo unico que me gustaria es que sean un poco ...	1
4	5	Great bang for the buck I can't believe the qu...	1

```

                                clean_text \
0                                happy with it
1 good quality material however the panels are m...
2 product was fun for bedroom windowsbr imagery ...
3 lo unico que me gustaria es que sean un poco ...
4 great bang for the buck i cant believe the qua...

```

```

                                tokens \
0                                [happy]
1 [good, quality, material, however, panels, mis...
2 [product, fun, bedroom, windowsbr, imagery, bi...
3 [lo, unico, que, gustaria, es, que, sean, un, ...
4 [great, bang, buck, cant, believe, quality, ma...

```

```

                                lemma \
0                                [happy]
1 [good, quality, material, however, panel, mism...

```

```

2 [product, fun, bedroom, windowsbr, imagery, bi...
3 [lo, unico, que, gustaria, e, que, sean, un, p...
4 [great, bang, buck, cant, believe, quality, ma...

```

```

                                preprocessed_text
0                                happy
1    good quality material however panel mismatched
2    product fun bedroom windowsbr imagery bit stre...
3    lo unico que gustaria e que sean un poco ma la...
4    great bang buck cant believe quality material ...

```

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
vectorizer = TfidfVectorizer(max_features=5000)
```

```
tfidf_features = vectorizer.fit_transform(df["preprocessed_text"])
```

```
tfidf_df = pd.DataFrame(
    tfidf_features.toarray(),
    columns=vectorizer.get_feature_names_out()
)
```

```
tfidf_df.head()
```

```

      10  100  1000  102  104  106  107  108  10884  108x108  ...  youre
youth  \
0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  ...  0.0
0.0
1  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  ...  0.0
0.0
2  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  ...  0.0
0.0
3  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  ...  0.0
0.0
4  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  0.0  ...  0.0
0.0

```

```

      youve  yr  yrsbecome  yuck  yuckjust  zen  zoom  zoomed
0      0.0  0.0          0.0  0.0          0.0  0.0  0.0  0.0
1      0.0  0.0          0.0  0.0          0.0  0.0  0.0  0.0
2      0.0  0.0          0.0  0.0          0.0  0.0  0.0  0.0
3      0.0  0.0          0.0  0.0          0.0  0.0  0.0  0.0
4      0.0  0.0          0.0  0.0          0.0  0.0  0.0  0.0

```

```
[5 rows x 5000 columns]
```

```
X = tfidf_df
```

```
y = df["label"]
```

```

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
LogisticRegression(max_iter=1000)
y_pred = model.predict(X_test)
y_pred
array([1, 1, 1, ..., 1, 1, 1], shape=(1060,))

from sklearn.metrics import accuracy_score
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy =", accuracy)
Accuracy = 0.9037735849056604

from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
print(cm)
[[ 24  97]
 [  5 934]]

from sklearn.metrics import classification_report
print(classification_report(y_test, y_pred))

```

	precision	recall	f1-score	support
0	0.83	0.20	0.32	121
1	0.91	0.99	0.95	939
accuracy			0.90	1060
macro avg	0.87	0.60	0.63	1060
weighted avg	0.90	0.90	0.88	1060

```
from sklearn.metrics import classification_report
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.83	0.20	0.32	121
1	0.91	0.99	0.95	939
accuracy			0.90	1060
macro avg	0.87	0.60	0.63	1060
weighted avg	0.90	0.90	0.88	1060

```
new_review = [
    "This product is amazing and worth buying."
]

new_review = [new_review[0].lower()]

new_review = [new_review[0].translate(str.maketrans('', '',
string.punctuation))]

tokens = word_tokenize(new_review[0])

tokens = [word for word in tokens if word not in stop_words]

tokens = [lemmatizer.lemmatize(word) for word in tokens]

processed_review = " ".join(tokens)

processed_review

'product amazing worth buying'

review_vector = vectorizer.transform([processed_review])

prediction = model.predict(review_vector)

if prediction[0] == 1:
    print("Positive Review 😊")
else:
    print("Negative Review 😞")
```

Positive Review 😊

C:\Users\gurug\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\LocalCache\local-packages\Python312\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
warnings.warn(